

Can your Data actually be forecasted?

Before choosing ARIMA, Prophet, or LSTM, ask yourself:

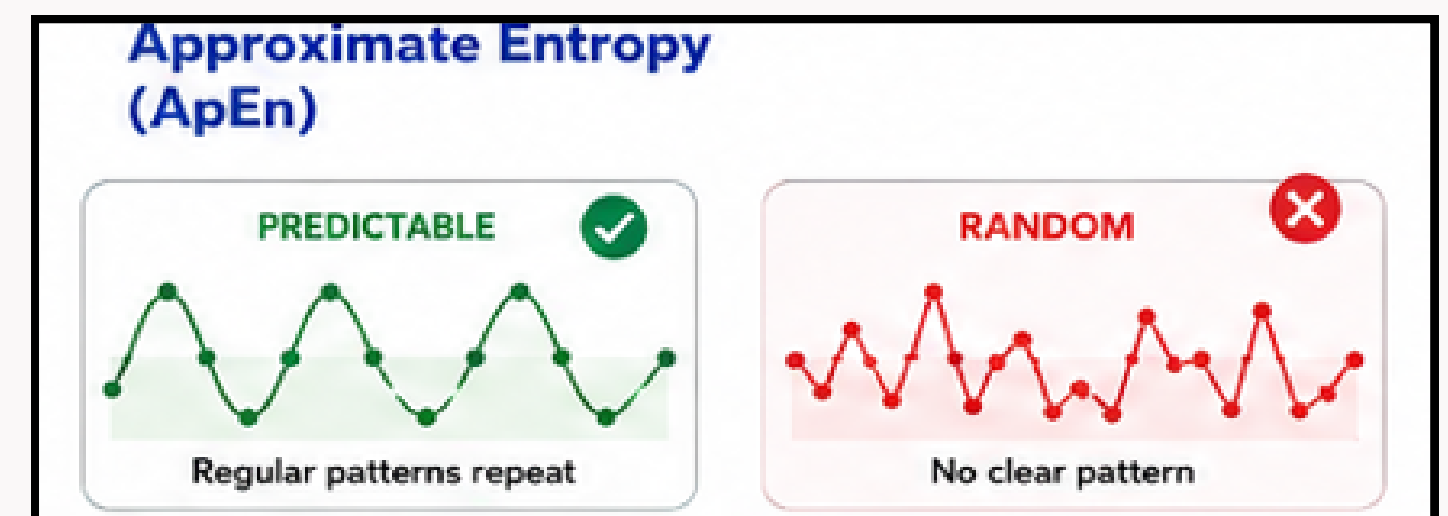
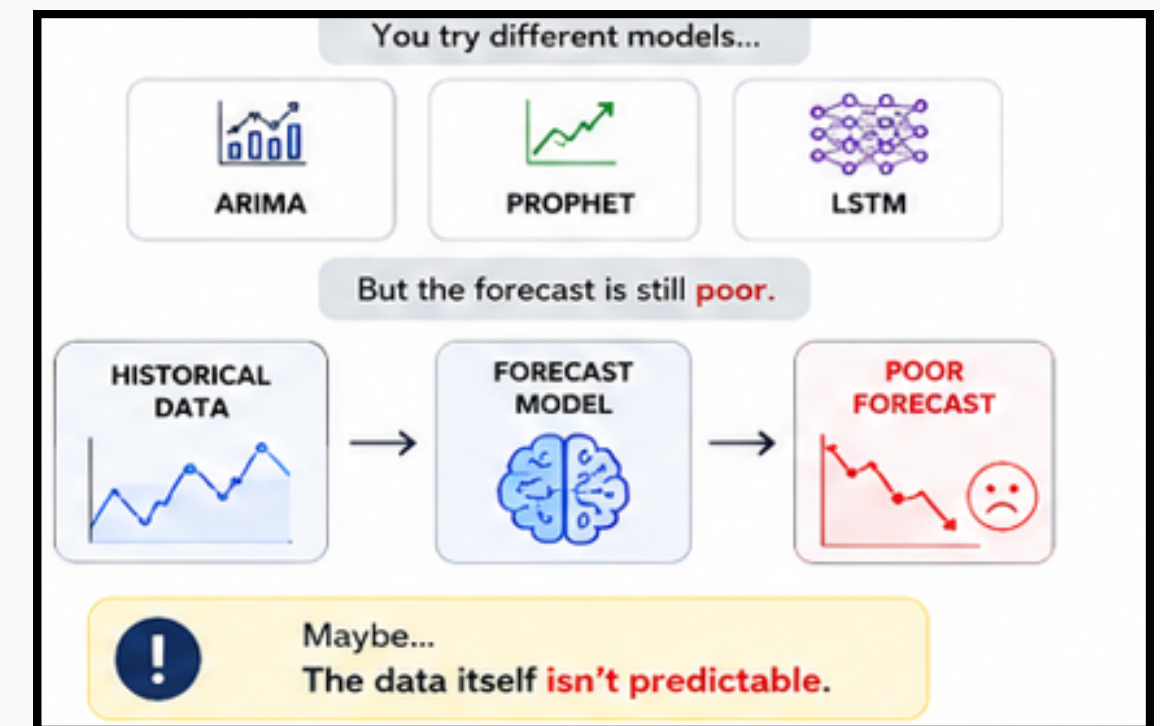
Is my data predictable enough to forecast?

One metric that helps answer this is:

Approximate Entropy (ApEn)

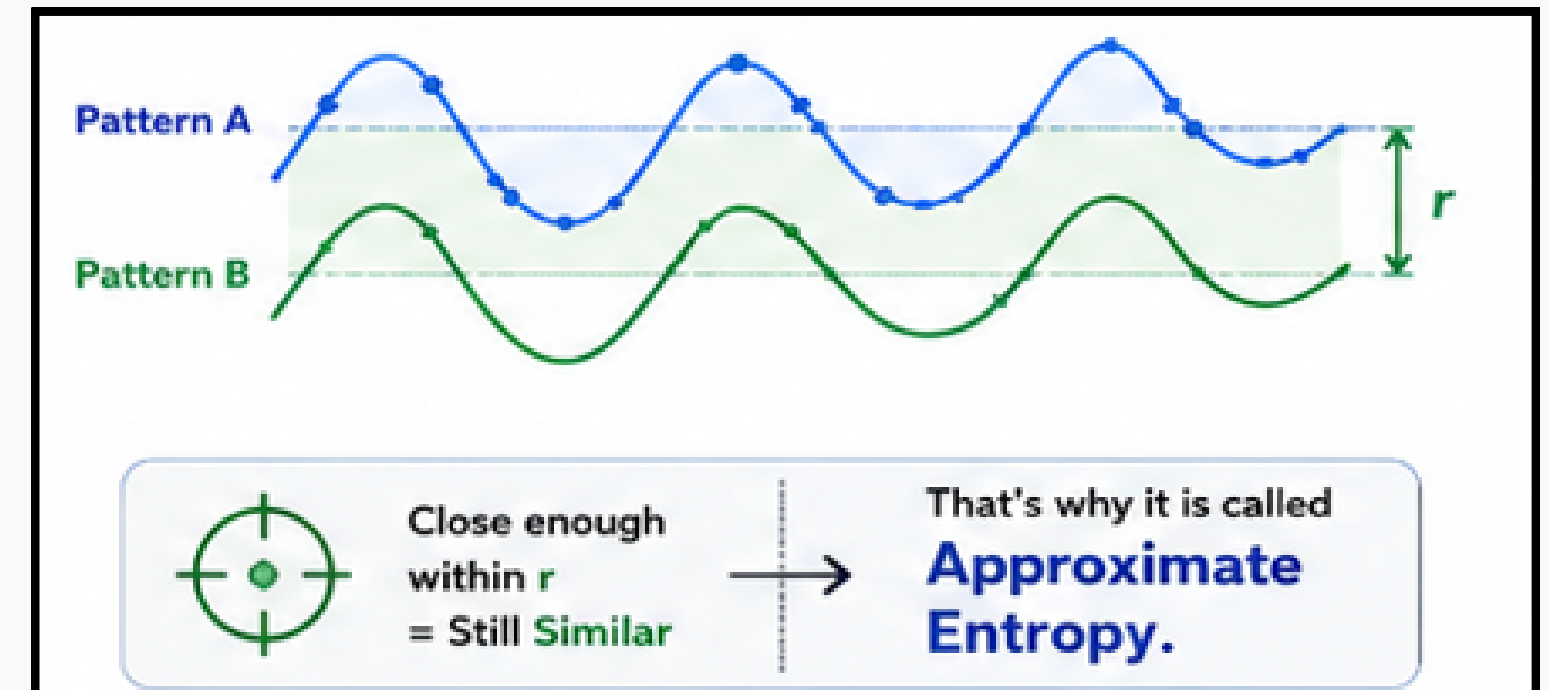
It measures how regular and predictable a time series is.

- Small ApEn → More predictable data
- Large ApEn → More random data



Why Approximate?

- Because the patterns don't have to be identical.
- They only need to be similar within a tolerance (r).
- The numbers are slightly different, but the overall trend is the same.
- That's why it's called Approximate Entropy



REAL WORLD EXAMPLE

Here we have taken 2 weeks of employee attendance records

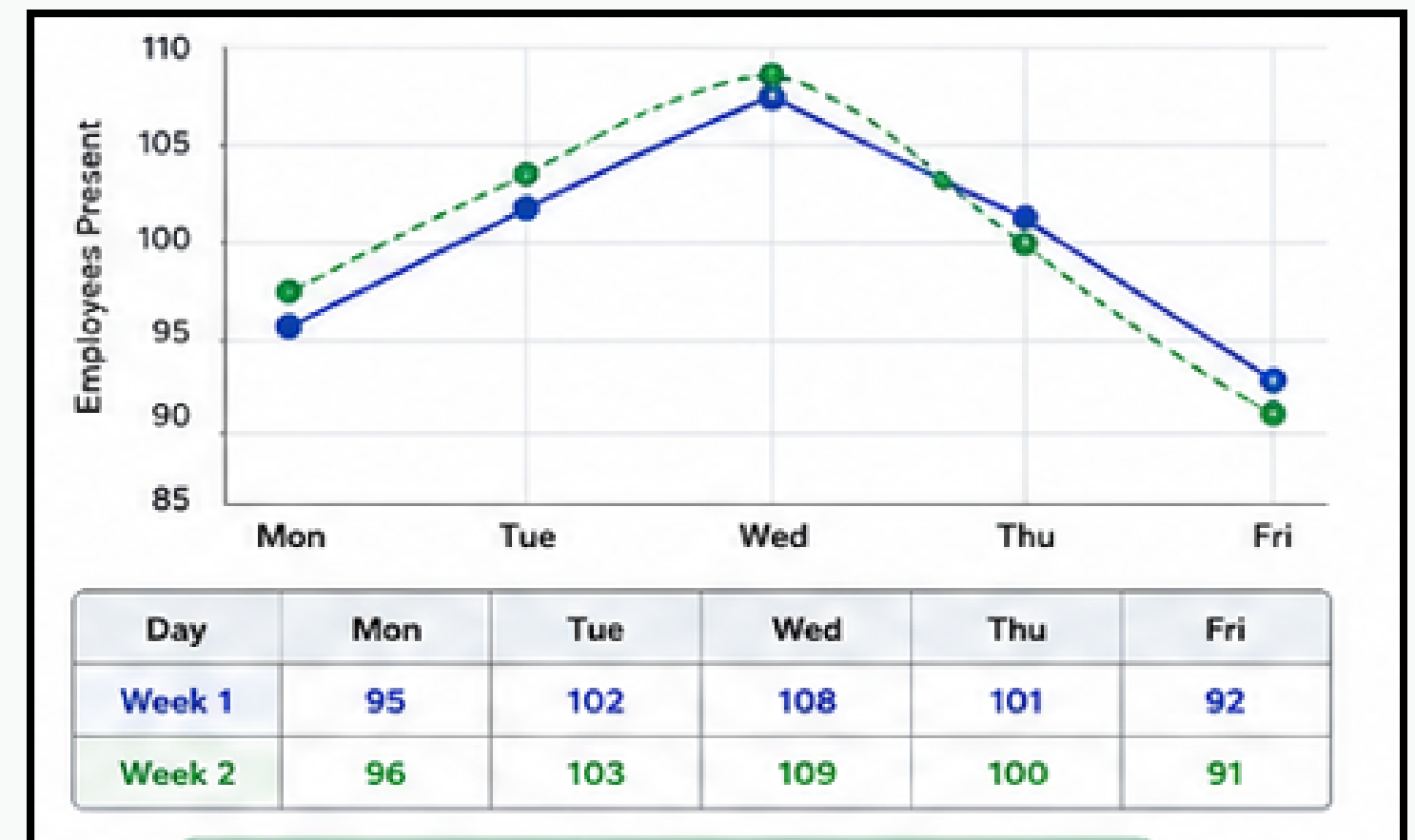
Notice the trend:



Attendance increases during the week.



Attendance drops on Friday.
The numbers are different, but the pattern is almost identical.



Why is the ApEn Small?

Does the overall pattern keep repeating?

Since the pattern is consistent,

→ Approximate Entropy is small.

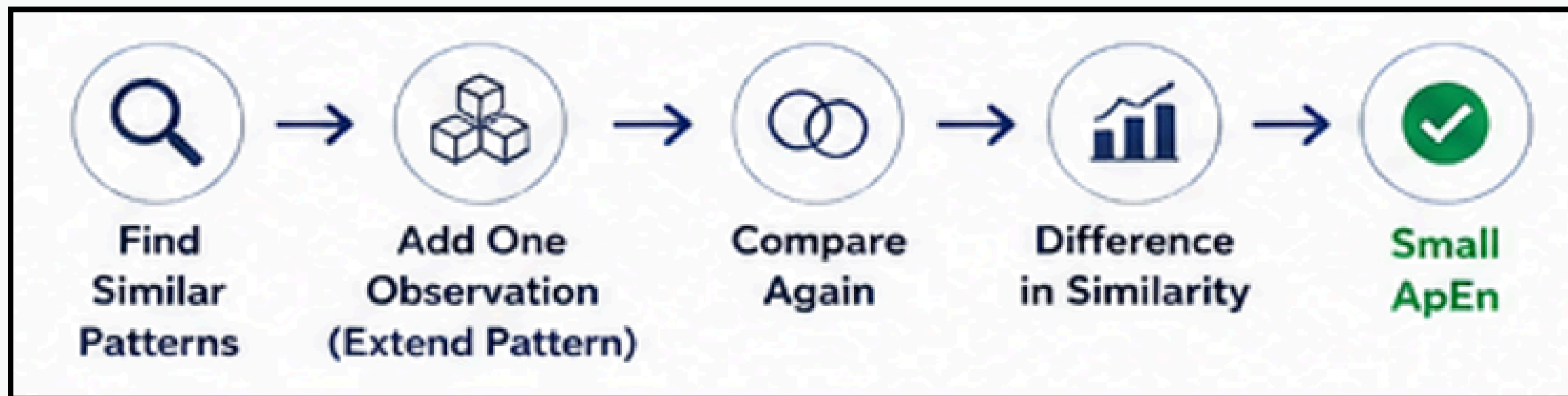


How ApEn is Calculated?

$$\text{ApEn}(m,r,N)=\phi(m)-\phi(m+1)$$

m: pattern length r: similarity tolerance

N: Number of Observations

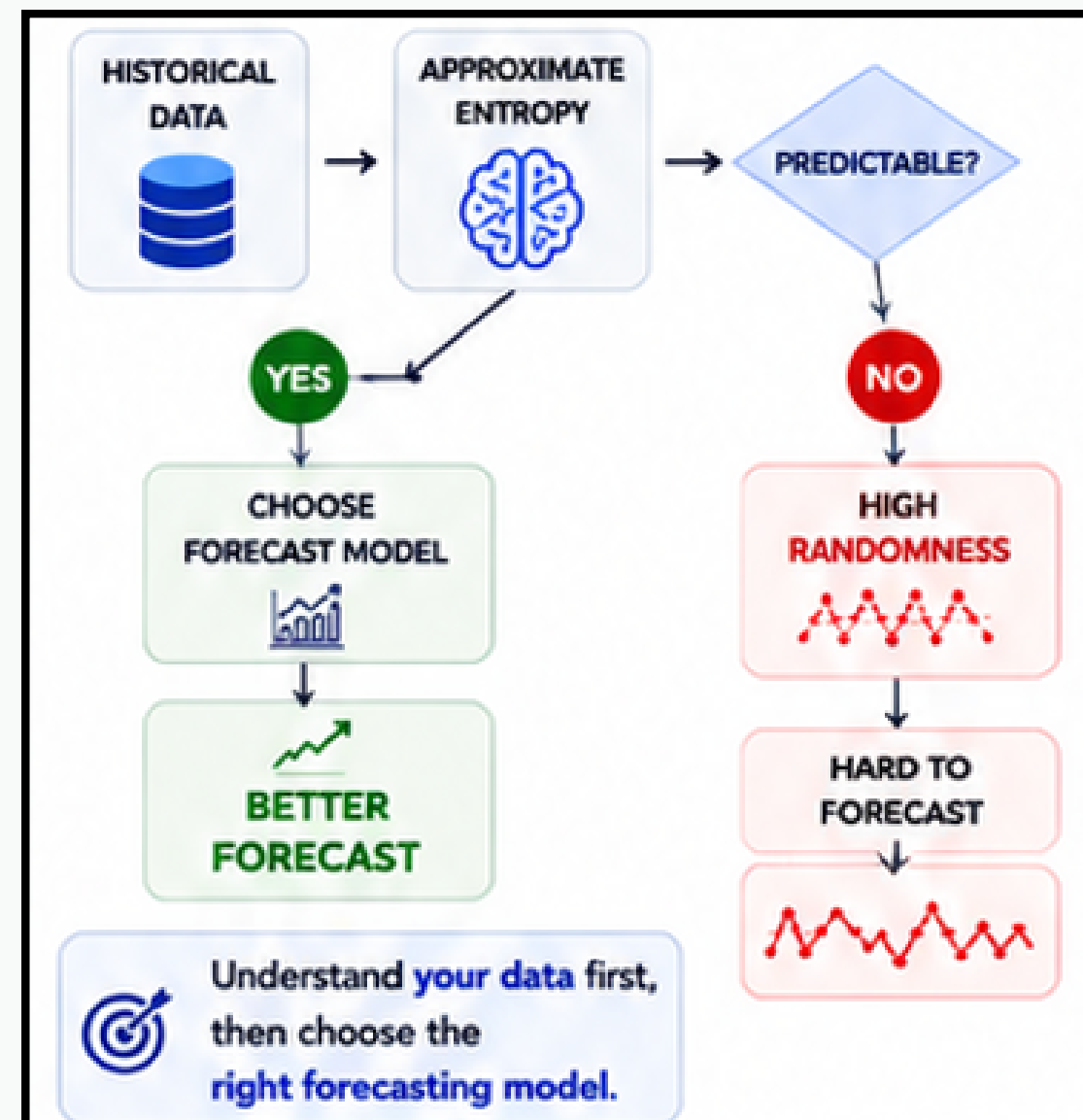


What does Small ApEn mean?

- The more a pattern repeats, the easier it becomes to predict.
- That's why repetitive data has a lower Approximate Entropy (ApEn).



Why should data Scientists Care?



Key TakeAway

Small **Approximate Entropy** indicates that the data exhibits consistent behaviour over time



When patterns are consistent, future values become easier to predict